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SUPERSEDING
Fed. Spec. W-B-411
December 31, 1941

FEDERAL SPECIFICATION

BLASTING MACHINE; GALVANOMETER, BLASTING; RESISTOR ASSEMBLY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers various types of hand-operated generators (blasting machines) used for detonating electric blasting caps or electric squibs (see sec. 6); one type of galvanometer used for checking the electrical blasting circuits for proper continuity; and one type resistor assembly used for testing blasting machines.

1.2 Classification.

1.2.1 *Groups and types.* The blasting apparatus covered by this specification shall be of the following groups and types, as specified (see 6.2):

Group A—Galvanometer for testing blasting circuits.

Group B—Blasting machines.
- Type I — 10 cap capacity.
- Type II — 30 cap capacity.
- Type III— 50 cap capacity.
- Type IV —100 cap capacity.

Group C.—Resistor assembly for testing blasting machines.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:
- TT-E-485—Enamel, Semi-gloss, Rust-Inhibiting.
- TT-E-489 — Enamel. Alkyd, Gloss (For Exterior and Interior Surfaces).
- TT-P-636 — Primer, Coating, Synthetic, Wood and Ferrous Metal.
- TT-P-664—Primer, Coating, Synthetic, Rust Inhibiting, Lacquer-Resisting.
- PPP-B-566—Boxes, Folding, Paperboard.
- PPP-B-585—Boxes, Wood, Wirebound.
- PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.
- PPP-B-601 — Boxes, Wood, Cleated-Plywood.
- PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.
- PPP-B-636—Box, Fiberboard.
- PPP-B-676—Boxes, Set-Up, Paperboard.
- PPP-T-76—Tape, Pressure Sensitive Adhesive, Paper, Water Resistant (For Carton Sealing).

Federal Standards:
- Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
- Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
- Fed. Std. No. 595—Colors.

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.

MIL-T-152 — Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.

MIL-V-173—Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic and Associated Electrical Equipment).

MIL-T-704—Treatment and Painting of Materiel.

MIL-L-10331—Leather, Moisture-and Mold-Resistant Treatment.

MIL-B-13136—Battery, Ba245, 2245.

MIL-S-13518—Sealer, Surface, Wood Preservative.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

MIL-STD-681—Identification Coding and Application of Hookup Wire.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the isue in effect on date of invitation for bids shall apply.

National Bureau of Standards:

Handbook H28—Screw-Threads Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

American Standards Association Standards:

ASA B6.1-1932—Spur Gear Tooth Form.

ASA B6.6-1946—Gear Tolerances and Inspection.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York 16, New York.)

Official Classification Committee Publication:

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

National Electrical Manufacturers' Association Standards:

NEMA Standards.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York 17, New York.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified, the contractor shall furnish complete blasting apparatus samples for preproduction inspection in accordance with 4.2 (see 6.2). Unless otherwise specified, one blasting apparatus sample shall be furnished for this inspection (see 6.2). The preproduction sample shall be identical with the proposed production items. The approved sample and the preproduction report shall be used as the production standard.

3.2 Design. The blasting apparatus shall be new. Group A shall be essentially a series type of ohmmeter for determining circuit conditions

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prior to firing of a single or a number of electric blasting caps or electric squibs in series. Group A shall consist of a milliameter in series with a suitable battery and resistance. Group B shall be essentially a manually operated generator capable of providing ample voltage for the detonation of a number of electric blasting caps or electric squibs in series. Group C shall consist of a number of resistance units that may be used singly or in combination for checking the output of blasting machines. All parts subject to wear, breakage, or distortion shall be accessible for adjustment and repair.

3.3 Materials. Materials shall be as specified herein. Materials not specifically designated shall be of good commercial quality. Contact between dissimilar metals, especially brass, copper, or steel in contact with aluminum or magnesium shall be avoided wherever possible.

3.4 Construction. The blasting apparatus shall be complete so that it can be used for the operations specified herein.

3.4.1 *Castings and forgings.* All castings shall be free from blowholes, porosity, hard spots, shrinkage, defects, cracks, or other defects. Forgings shall be free from scale, inclosures, cold shuts, mismatching and other defects that may affect the structural strength of the forgings involved. Strength and other essential physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein.

3.4.2 *Welding, brazing and soldering.* Welding, brazing, and soldering shall be performed in accordance with good commercial practice. In no event shall such processes, including peening or plugging, be used on castings or forgings for reclaiming any parts of the blasting apparatus without authorization of the procuring activity.

3.4.3 *Fastening devices.* All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. All such parts, when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.4 *Standard parts.* MIL Standard parts shall be used where applicable. In event MIL Standard parts are not available, commercial utility parts such as screws, nuts, bolts, etc., may be used provided they conform to the requirements of this specification.

3.4.5 *Electrical connections.* Electrical connections shall be mechanically secured and soldered, or brazed; except connections on screw type terminals which shall be mechanically secured.

3.4.6 *Threads.* All threaded parts shall be in the inch system and shall comply with National Bureau of Standards Handbook H28. Unified screw threads shall be utilized wherever possible in lieu of American National screw threads and nonunified American Standard screw threads.

3.4.7 *Gears.* All gears and pinions shall be fabricated in the English system and shall conform to American Standards Association Inc. Standard B6.6. If tolerances other than those specified are required to manufacture gears or pinions, such tolerances shall assure accuracy of operation of the blasting apparatus equal to or exceeding that specified herein. Spur-gear tooth form shall conform to American Standards Association Inc. Standard B6.1.

3.4.8 *Lubrication.* All bearings (except sealed-for-life type) and other working parts shall be provided with means for lubrication. Apparatus shall be lubricated wherever necessary prior to delivery.

3.4.9 *Safety devices.* Each blasting apparatus shall be furnished with safety devices. Parts which are hazardous to the operator shall be guarded. Protection against electric shock shall be provided. When specified, special safety devices shall be provided as required by the procuring activity (see 6.2).

3.4.10 *Interchangeability.* All replaceable parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type of model can be replaced or adjusted without requiring modi-

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fication. Where practicable, all such parts shall be permanently and legibly marked with the manufacturer's part number in accordance with Military Standard MIL–STD–130.

3.4.11 *Surfaces.* All castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smoothed and all edges shall be rounded or beveled.

3.4.12 *Protective finishes.* Exposed metal and working surfaces except when parts are inherently corrosion resistant, shall be finished to resist corrosion. Exposed metal is defined as metal exposed to atmospheric conditions.

3.5 Performance and product characteristics.

3.5.1 *Group A.* The galvanometer shall be a ruggedized, portable instrument and shall be capable of producing a full-scale deflection of the indicator from a load current of not more than 8 milliamperes.

3.5.1.1 *Position influence.* Position influence shall not exceed 1 percent of the readings obtained in the normal position.

3.5.1.2 *Range of zero adjustment.* A minimum adjustment of 3 percent of full-scale length shall be provided.

3.5.2 *Group B.* The blasting machine shall be of sturdy construction, capable of withstanding rough handling and usage in the field. The machines shall be capable of firing standard electric blasting caps with the current output specified herein. Type I blasting machine shall be capable of withstanding a minimum of 5000 actuations without damage; types II, III, and IV a minimum of 2500 actuations without damage.

3.5.2.1 *Type I.* The blasting machine shall have a capacity of firing 10 electric blasting caps or electric squibs in series. The blasting machines shall deliver at least 1.5 amperes through a total external circuit resistance of 32 ohms; and shall deliver at least one ampere for a period of not less than 10 milliseconds.

3.5.2.2 *Type II.* The blasting machine shall have a capacity of firing 30 electric blasting caps or electric squibs in series. The blasting machines shall deliver at least 1.5 amperes through a total external circuit resistance of 96 ohms; and shall deliver at least one ampere for a period of not less than 10 milliseconds.

3.5.2.3 *Type III.* The blasting machine shall have a capacity of firing 50 electric blasting caps or electric squibs in series. The blasting machine shall deliver at least 1.5 amperes through a total external resistance of 160 ohms; and shall deliver at least one ampere for a period of not less than 10 milliseconds.

3.5.2.4 *Type IV.* The blasting machine shall have a capacity of firing 100 electric blasting caps or electric squibs in series. The blasting machine shall deliver at least 1.5 amperes through a total external resistance of 320 ohms; and shall deliver at least one ampere for a period of not less than 10 milliseconds.

3.5.3 *Group C.* Resistor assembly shall be capable of providing ohmic resistance values with tolerances of plus or minus 5 percent.

3.5.4 *Dimensions and weight.* The blasting apparatus shall conform to the following dimensions and weight:

GROUP	TYPE	HEIGHT (Maximum)	WIDTH (Maximum)	DEPTH (Maximum)	WEIGHT (Maximum)
A	(Less carrying case)	4¼ in.	3¼ in.	1¾ in.	1 Pound w/o battery
B	I	8 in.	4 in.	3 in.	6 Pounds
B	II	16 in.	8 in.	8 in.	25 Pounds
B	III	17 in.	8 in.	8 in.	28 Pounds
B	IV	18 in.	8 in.	10 in.	32 Pounds
C		2¾ in.	8¼ in.	1½ in.	1 Pound

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3.5.5 *Temperature and humidity conditions.*

3.5.5.1 *Storage temperatures.* The blasting apparatus and all components thereof shall be able to withstand storage without damage in a temperature of minus 65°F. for periods up to 72 hours, and a 24 hour cycle of temperature, starting with 10 hours at plus 90°F., followed by 5 hours of steady increase to plus 155°F., 4 hours at plus 155°F., and 5 hours of steady decrease to plus 90°F. Following these extreme exposures and when stabilized at plus 80°F. ± 10°F., the blasting apparatus shall conform to the performance requirements of 3.5.1, 3.5.2 or 3.5.3 as applicable.

3.5.5.2 *Operating temperature.* The blasting apparatus and all components thereof shall be able to withstand without damage and operate within the performance requirements set forth in 3.5.1, 3.5.2 or 3.5.3, whichever is applicable, while being subjected to an operating temperature of minus 40°F. and plus 125°F.

3.5.5.3 *Humidity conditions.* The blasting apparatus shall operate within the performance requirements of 3.5.1, 3.5.2 or 3.5.3, whichever is applicable, after exposure to 20 hours of relative humidity between 93 and 97 percent at air temperatures from plus 80°F. to plus 85°F., followed by 4 hours of 100 percent relative humidity with condensation at plus 75°F. to plus 80°F. Air movement and radiation negligible.

3.5.6 *Vibration.*

3.5.6.1 *Group A.* When the galvanometer is subjected to simple harmonic motion for 1 hour in each of the three major axial directions at an amplitude of 0.030 inch (0.060 in total excursion) and at a frequency variation such that the entire range between 10 and 55 cycles per second is covered each minute, the galvanometer shall show no damage, and the permanent change in indication shall be not greater than 0.5 percent of the readings prior to vibration.

3.5.6.2 *Group B and group C.* The blasting apparatus shall be capable of being subjected to a vibration of simple harmonic motion, for 1 hour in each of the three major axial directions, at an amplitude of 0.030 inch (0.060-inch total excursion) and at a frequency variation such that the entire range between 10 and 55 cycles per second is covered each minute. Following vibration, the blasting apparatus shall show no damage and shall conform to the performance requirements of 3.5.2 or 3.5.3 as applicable.

3.5.7 *Shock.*

3.5.7.1 *Group A.* The galvanometer shall withstand, without damage, being dropped a vertical distance of 8 inches, through each of its three major axes, on a concrete surface. Permanent change in indication shall be not greater than 0.5 percent of the readings prior to being subjected to shock.

3.5.7.2 *Group B and Group C.* The blasting apparatus shall withstand, without damage, being dropped a vertical distance of 4 feet through each of its major axes on a concrete surface. Following, the blasting apparatus shall conform to the performance requirements of 3.5.2 or 3.5.3.

3.5.8 *Dielectric strength.*

3.5.8.1 *Group A and Group C.* The blasting apparatus shall be capable of withstanding an application of twice rated voltage applied for one minute between mutually insulated circuits and between insulated circuits and external metal parts. Following, the blasting apparatus shall conform to the performance requirements of 3.5.1 or 3.5.3, as applicable.

3.5.8.2 *Group B.* The blasting machines shall be capable of withstanding an application of a high potential applied for one minute between mutually insulated circuits and between insulated circuits and external metal parts. Potentials applied shall be 200 volts for type I, 600 volts for type II, and 1000 volts for type III and type IV. Following, the blasting machines shall conform to the performance requirements of 3.5.2.

3.5.9 *Water submersion (group B).*

3.5.9.1 *Type I.* Type I blasting machine shall be capable of being subjected to submer-

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sion of 8 hours in 4 feet of water at approximately the same temperature as that of the machine at the time of immersion. Following the submersion, the blasting machine shall conform to the performance requirements of 3.5.2.1.

3.6 Detail of components. The blasting apparatus shall consist of the following principal component parts:

3.6.1 *Type I blasting apparatus.*

(*a*) Meter indicator

(*b*) Meter case

(*c*) Carrying case

3.6.1.1 *Meter indicator.* The instrument shall be basically a milliameter in series with a suitable resistance.

3.6.1.1.1 *Scale.* The scale plate shall be lithographed in black characters on a fine frosted silver or etched background or on a lithographed white background. The scale shall have 25 mechanically uniform divisions, each fifth line being wider than the others and extending below the lower arc. The extended lines shall be figured 0, 5, 10, 15, 20, and 25 in order. The figure 25 shall indicate full-scale deflection. The figures shall be not less than 0.10 inch high. The cardinal lines shall be not less than 0.15 inch high. The intermediate lines shall be not less than 0.08 inch high. An additional scale calibrated in ohms and not superimposed upon the uniform scale may be provided. The length of the scale shall be not less than 1¾ inches as measured by the travel of the tip of the pointer.

3.6.1.1.2 *Pointer.* The pointer shall be of aluminum alloy and shall have a lance or pear-shaped tip. The pointer shall be of such dimensions and shape as to provide sufficient rigidity for portable instruments subjected to rugged field use. The visible length of the pointer shall be coated a dull black. The tip shall extend above the lower arc of the scale.

3.6.1.1.3 *Zero adjustment.* An external zero adjustment shall be provided to enable setting the pointer to zero without removing the instrument from the carrying case.

3.6.1.1.4 *Resistance unit.* A resistor, or resistors, shall be provided in series with the battery source and milliameter and shall be of such resistance that a new silver chloride battery having a measured voltage of 0.90 volts (see 3.6.4.1), will just bring the pointer to full scale deflection when the contact studs are short circuited. The resistor, or resistors, shall be mounted within the indicator compartment.

3.6.1.1.5 *Contact studs.* Two nickel plated brass contact studs 7/32 inch long and 3/8 inch in diameter shall be provided as the outlets for the meter circuit. The top surfaces shall be serrated in order to provide good pressure contact when the instrument is in use.

3.6.1.2 *Meter case.* The galvanometer shall be contained in a durable rectangular case made of metal, phenolic composition, or other suitable plastic material. Plastics having a cellulose nitrate base are not acceptable. The interior of the case shall have two compartments; an indicator compartment and a bateery compartment. The construction shall be such as to resist the passage of moisture, fumes, or dust between the two compartments or between the atmosphere and either of the two compartments. The indicator compartment shall be sealed to prevent or indicate tampering. The battery compartment shall be readily accessible for the purpose of replacing the silver chloride dry cell battery.

3.6.1.3 *Meter carrying case.* The galvanometer shall be inclosed in a durable carrying case not less than 5/64 inch thick. A carrying strap not less than 5/64 inch thick, ½ inch wide and at least 4 feet long shall be securely fastened to the carrying case and shall be provided with a buckle for adjusting the length of the carrying strap. The carrying case shall have an opening or openings for the two contact studs. An opening of such dimensions shall be provided so that the scale readings and the movement of the tip of the pointer across the scale shall be visible. Access opening to the zero adjustment shall be provided. If a flap is provided which normally conceals any or all openings, provisions shall be made so that when the flap is opened the galvanometer cannot fall out of the carrying case even when turned upside down.

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3.6.2 *Group B blasting apparatus.*
(*a*) Generator
(*b*) Operating mechanism
(*c*) Enclosure

3.6.2.1 *Type I.*

3.6.2.1.1 *Generator.* The generator of the type I blasting machine shall be of the self-excited, shunt or compound wound direct-current type. The armature windings and field coils shall be carefully insulated and impregnated with suitable varnish or compound whose melting point shall be above 80°C. Neither shellac nor paraffin shall be used for impregnating windings or coils. The material and construction of the magnetic circuit shall be such as to retain a strong residual magnetism.

3.6.2.1.1.1 *Switch.* A suitable and durable switch shall be provided and connected electrically so that the accumulated energy of the generator is delivered into the electric blasting cap circuit at or near the end of the operating stroke.

3.6.2.1.2 *Operating mechanism.* The operating spindle shall extend through the top of the enclosure and shall be operated by a suitable detachable handle. The protruding portion of the operating spindle shall be enclosed within a shoulder or collar so designed that the operating spindle cannot be operated except by the handle. The operating handle shall be proportioned and dimensioned so as to permit the operator a firm grasp without striking the knuckles during the twisting or turning movement. Proper function shall be accomplished with one twist of the handle, a maximum of 120 degrees. The operating spindle shall actuate a train of gears to revolve the armature of the generator at an adequate speed and for the proper time duration for the detonation of electric blasting caps as specified herein. White metal shall not be used for gears or pinions.

3.6.2.1.3 *Enclosure.* The enclosure shall be a rugged, durable, waterproof metal case. Provision for attaching the operating handle to the enclosure shall be provided to prevent loss.

3.6.2.1.3.1 *Carrying strap.* A carrying strap not less than 1 inch wide and not less than $\frac{5}{64}$ inch thick shall be fastened securely to the body of the blasting machine in such a manner that it can also be used as a stirrup on the under side of the body of the machine, through which the operator may slip his hand for assistance in holding the machine firmly in one hand while operating the key handle with the other.

3.6.2.1.3.2 *Binding posts.* Two binding posts with suitable knurled thumb nuts that cannot be removed shall be mounted on the top of the case, so proportioned as to permit wires as small as 0.032 inch in diameter and as large as 0.081 inch in diameter to be firmly clamped by the thumb nuts. Binding posts and thumb nuts shall be nonferrous metal.

3.6.2.2 *Types II, III, and IV.* With the exception of cap capacity, requirements shall be the same for types II, III, and IV.

3.6.2.2.1 *Generator.* The generator shall be totally enclosed and of the direct-current type. The generator shall be provided with bearing plates at each end which shall completely enclose the generator windings. The end plates or end bells of the generator shall be fitted with high grade removeable bearing bushings of suitable material and composition. One end bearing plate shall be suitable for carrying the brushholders and the other bearing plate shall be provided with a channeled guide for the rack bar. The armature windings and field coils shall be carefully insulated and impregnated with suitable varnish or compound whose melting point shall be above 80°C. The material and construction of the magnetic circuit shall be such as to retain a strong residual magnetism.

3.6.2.2.1.1 *Switch.* A suitable and durable switch shall be provided which will operate by the movement of the rack bar at or near the end of the stroke and which is so connected electrically that the accumulated energy of the generator is delivered to the electric blasting cap circuit at or near the end of the stroke of the rack bar.

3.6.2.2.2 *Operating mechanism.* A rack shall extend through the top of the enclosing box and shall have a wood handle for actuating the machine by a forceful downward stroke. The

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wood handle of the rack bar shall be of such shape and size as to readily permit the operator to use both hands in the most efficient and practical manner. The gears and the pinions comprising the train from the rack bar to the armature shaft shall be of such material and strength as to withstand the hard service and rough usage likely to occur in service. White metal shall not be used for gears or pinions. The rack bar shall be supported in proper mesh and alignment and shall be independent of the containing box. Proper function of the machine shall be accomplished with one downward stroke of the handle. The entire gear train and rack bar shall be an independent and operable unit and shall revolve the armature of the generator at an adequate speed and for the proper time duration for the detonation of electric blasting caps as specified herein.

3.6.2.2.3 *Enclosure.* The enclosing box shall be made of selected plain oak or ashwood, fully seasoned. Each panel and the top and bottom of the box shall be of one piece. Two or more pieces glued or otherwise joined are not acceptable. The box shall be so constructed that at least two opposite sides are readily removable for inspection of the blasting machine. The shelf supporting the generator shall be readily removable for purposes of inspection or repairs. The four sides of the box shall be not less than ½ inch thick, the top and bottom not less than $^{13}/_{16}$ inch thick. The upper and lower edges of the sides shall be dovetailed to the top and bottom pieces. The interior surfaces of the box and the surfaces of the shelf shall be painted, sprayed or dipped with a moisture resisting paint or varnish. The exterior of the box shall be sealed in conformance with the Group A requirements of Military Specification MIL–S–13518.

3.6.2.2.3.1 *Carrying strap.* A carrying strap not less than 1 inch wide and not less than ⅛ inch thick shall extend across the top of the enclosure and shall be fastened to the sides of the box in such a manner as to be easy to remove and replace.

3.6.2.2.3.2 *Binding posts.* Two binding posts shall be provided with suitable knurled thumb nuts that cannot be removed. The binding posts shall be rigid and strong enough to withstand any pressure or twisting by hand force. They shall be doweled or so attached to the wood box that they cannot twist or turn or become loosened through normal use. The binding posts shall be so designed as to firmly clamp wires as small as 0.032 inch in diameter and as large as 0.102 inch in diameter. Binding posts and thumb nuts shall be of nonferrous metal.

3.6.3 *Group C, blasting apparatus.*

(*a*) Resistor assembly.

(*b*) Resistance units.

3.6.3.1 *Resistor-assembly case.* The resistor assembly shall have a substantial case of molded bakelite, oak, ash, or equal hardwood and a panel of suitable plastic composition on which the binding posts and resistance units shall be mounted.

3.6.3.1.1 *Binding posts.* The number of binding posts furnished shall be adequate for the resistance unit requirements specified in 3.6.3.2. The panel shall be marked in numbers denoting the number of equivalent electric blasting caps substituted in series when using the various binding posts. The binding posts shall be made of nickel-plated brass and shall be of such a type that the thumb nut cannot be removed. They shall be of adequate dimensions to accommodate wire up to 0.064 inch in diameter. The binding posts with the thumb nut in the raised position shall not extend for more than 1 inch above the top of the case. All connections shall be securely soldered.

3.6.3.2 *Resistance units.* The value of the individual resistance units between consecutive binding posts shall be as follows:

Number of caps or squibs in series	Resistance in Ohms
5	16
10	32
20	64
25	80
40	128

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Sufficient current carrying capacity shall be provided for any setting to prevent damage to the resistor assembly by momentary current supplied by a blasting machine.

3.6.4 *Electrical equipment.*

3.6.4.1 *Group A, power requirement.* Unless otherwise specified (see 6.2), the galvanometer shall be capable of operating from a silver chloride battery MIL Type BA–245/U in accordance with Military Specification MIL–B–13136. The battery shall not be supplied with the galvanometer.

3.6.4.2 *Electrical components.* Electrical components of the blasting apparatus including switches, wiring and their enclosures shall conform to and shall be located in accordance with the NEMA Electrical Standards, except as specified herein. No component parts shall be used in any application where the requirements exceed the manufacturer's rating for the part. Internal wiring shall be color coded in accordance with Military Standard MIL–STD–681.

3.7 Fungus treatment. The blasting apparatus shall be treated for fungus resistance in accordance with the requirements of Military Specification MIL–T–152, using material conforming to requirements of Military Specification MIL–V–173. Leathers shall be treated for moisture and mold resistance in accordance with the requirements of Military Specification MIL–L–10331.

3.8 Painting. Unless otherwise specified (see 6.2), painting and finishing of the Group B blasting apparatus shall be as follows:

3.8.1 *Preparation of surface.* All surfaces to be painted shall be cleaned of all grease, oil, scale, rust, dirt, casting sand, slag, and other extraneous matter and shall be dry.

3.8.2 *Application.* Application shall be in accordance with the requirements of Military Specification MIL–T–704, except that the Group B blasting apparatus shall be finished with one coat of primer conforming to the requirements of Federal Specification TT–P–636 or TT–P–664, followed by two coats of enamel conforming to the requirements of Federal Specification TT–E–489. For Navy purchases, the primer shall conform to Federal Specification TT–E–485 and shall require the application of only one finish coat. Color shall be No. 16187 (gray) as shown in Federal Standard No. 595.

3.9 Identification of product. The blasting apparatus shall be marked for identification in accordance with Military Standard MIL–STD–130.

3.10 Instruction plates.

3.10.1 *Group A.* A corrosion resistant instruction plate shall be attached to the case bearing a warning to the effect that the battery used must be a silver chloride dry cell type and, indicating when more than one cover is provided, the location of the battery cover.

3.10.2 *Group B.* A brass instruction plate shall be attached to the outside of the case of each blasting machine giving the rated capacity expressed in the number of electric blasting caps, each having copper leg wires 30 feet in length, that can be fired in series, and the proper method of operating the unit. On type I blasting machines, the instruction plate shall include the notation: "This machine may lose capacity temporarily; to restore, twist handle vigorously a few times."

3.10.3 *Group C.* A brass instruction plate having directions for the use of this instrument plainly stamped or etched thereon shall be securely attached to the case.

3.11 Handbook of instructions. Unless otherwise specified, the contractor shall furnish two copies of a handbook of instructions with each blasting apparatus supplied under this specification. The handbook shall contain necessary operating instructions, drawings, testing instructions, illustrations, parts list, manufacturer's part numbers, lubrication instructions, assembly and disassembly instructions and safety precautions, to insure proper operation and maintenance of the blasting apparatus (see 6.2). For the Group A blasting apparatus, a handbook

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shall be placed within the carrying case. Additional copies of the instruction handbook and the mailing address to which additional copies shall be sent will be specified by the procuring activity. (See 6.2.)

3.11.1 *Handbook approval.* When specified, a preliminary copy of the instruction handbook shall be submitted to the contracting officer for approval prior to publication and delivery. (See 6.2.)

3.12 Workmanship. Workmanship shall be first class to assure procurement of good quality equipment of a neat general appearance. Standards of workmanship shall be such that equipment specified herein shall comply with the requirements in section 3. All components, subassemblies, parts, and the final assembly shall comply with tests in section 4.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. Unless otherwise specified, preproduction inspection shall consist of all the tests specified in 4.4 and examination in accordance with 4.3.1 except that sampling in accordance with Military Standard MIL–STD–105 shall not apply (see 6.2). Preproduction samples shall be as specified in 3.1. Unless otherwise specified, preproduction inspection shall be performed by the contractor (see 6.2). Failure of preproduction sample to pass preproduction inspection shall be cause for rejection.

4.3 Acceptance inspection. Acceptance inspection shall consist of the examination specified in 4.3.1 and the tests specified in 4.3.2.

4.3.1 *Acceptance examination.* The blasting apparatus shall be examined in accordance with Military Standard MIL–STD–105 using the following classification of defects and acceptable quality level (AQL):

4.3.1.1 *Classification of defects.*

Major Defects

(AQL shall be 2.5 percent defective)

101. Failure of unit to operate (see 3.4).
102. Castings and forgings not as specified (see 3.4.1).
103. Defective welding, brazing, or soldering affecting operation (see 3.4.2).
104. Defective or insecure fastening devices affecting operation (see 3.4.3).
105. Safety devices unsatisfactory and special safety devices not furnished when specified (seee 3.4.9).
106. Replacements parts not interechangeable or marked as specified (see 3.4.10).
107. Dimensions and weight not as specified (see 3.5.4).
108. Omission of a component part (see 3.6.1, 3.6.2 and 3.6.3).
109. Meter scale not as specified (see 3.6.1.1.1).
110. Pointer not as specified (see 3.6.1.1.2).
111. Meter case not as specified (see 3.6.1.2).
112. Generator not as specified (see 3.6.2.1.1 and 3.6.2.2.1).
113. Operating mechanism not as specified (see 3.6.2.1.2 and 3.6.2.2.2).
114. Resistance units not as specified (see 3.6.3.2).

Minor Defects

(AQL shall be 6.5 percent defective)

201. Threads not as specified (see 3.4.6).
202. Meter carrying case not as specified (see 3.6.1.3).
203. Enclosures not as specified (see 3.6.2.1.3 and 3.6.2.2.3).
204. Resistor assembly case not as specified (see 3.6.3.1).

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205 Painting and finishing not as specified (see 3.8).
206 Identification marking not as specified (see 3.9).
207 Instruction plates not as specified (see 3.10).
208 Handbook of instructions omitted or incomplete (see 3.11).

4.3.2 *Acceptance tests.* From the samples which have passed the acceptance examination specified in 4.3.1, samples for acceptance tests shall be selected in accordance with inspection level L5 of Military Standard MIL-STD-105. The samples selected shall be subjected to the performance tests (see 4.4.1.1, 4.4.1.2, and 4.4.1.3). Acceptance or rejection shall be based on an AQL of 4.0 percent defective.

4.4 Test methods.

4.4.1 *Performance tests.*

4.4.1.1 *Group A.* Tests shall be conducted in accordance with instructions described in the handbook furnished with each galvanometer. Methods of measurement shall have an accuracy error of not more than ¼ percent.

4.4.1.1.1 *Position influence test.* Position influence is defined as the maximum displacement of the pointer which is caused solely by a 60° inclination of the meter from normal operating position in the direction which produces the most unfavorable conditions as to position. Normal operating position of the meter is when the galvanometer is resting on the bottom of the case, with the meter window in a horizontal plane. With the galvanometer loaded to produce half scale deflection, position influence shall not exceed 1 percent of the readings obtained in the normal position. Failure to meet this test shall be cause for rejection of the galvanometer (see 3.5.1.1).

4.4.1.1.2 *Zero adjustment test.* The galvanometer shall be tested to determine that it is provided with a minimum adjustment of 3 percent of full-scale length. Failure to meet this test shall be cause for rejection of the galvanometer (see 3.5.1.2).

4.4.1.2 *Group B.* Tests shall be conducted in sequence as specified herein; the life cycle test being performed prior to the current output test.

4.4.1.2.1 *Life-cycle test.* The type I blasting machine shall be operated a total of 5000 times at a minimum rate of 20 actuations per minute. The type II, III, and IV machines shall be operated a total of 2500 times at a minimum rate of 20 actuations per minute. Upon completion of this test, the blasting machine shall be completely disassembled and all parts examined. Any evidence of wear or damage or failure to meet the current output requirements of 4.4.1.2.2 following reassembly, shall constitute failure of the test (see 3.5.2).

4.4.1.2.2 *Current output tests.* The blasting machine shall be tested to determine its capability of delivering at least 1.5 amperes through a total external circuit resistance of 32 ohms for type I, 96 ohms for type II, 160 ohms for type III and 320 ohms for type IV. The blasting machine shall also be tested to determine its capability of delivering at least one ampere for a period of not less than 10 milliseconds. Failure to deliver the currents specified herein shall be cause for rejection of the blasting machine. (See 3.5.2.1, 3.5.2.2, 3.5.2.3 and 3.5.2.4.)

4.4.1.3 *Group C.* The resistor assembly shall be tested for its capability of providing ohmic resistance values between consecutive binding posts as follows:

Number of caps or squibs in series	Ohmic resistance
5	16
10	32
20	64
25	80
40	128

Failure to met these ohmic resistance values within tolerances of plus or minus 5 percent shall be cause for rejection of the resistor assembly. (See 3.5.3.)

4.4.2 *Storage tests.*

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4.4.2.1 *Low temperature test.* The blasting apparatus shall be placed in an ambient air temperature of minus 60°F. to minus 65°F. for a period of 72 hours. At the end of this time, the blasting apparatus shall be removed and allowed to stabilize at an air temperature of plus 80°F. ± 10°F. The blasting apparatus shall then be inspected for evidence of damage and tested for performance in accordance with 4.4.1.1, 4.4.1.2.2 and 4.4.1.3. (See 3.5.5.1.)

4.4.2.2 *High temperature test.* The blasting apparatus shall be submitted to a 24 hour cycle of temperature starting with 10 hours at plus 90°F., ±5°F., followed by 5 hours of steady increase to plus 150°F. to plus 155°F., 4 hours at plus 150°F. to plus 155°F., and 5 hours of steady decrease to plus 90°F., ±5°F. The blasting apparatus shall then be inspected for evidence of damage and tested for performance in accordance with 4.4.1.1, 4.4.1.2.2 and 4.4.1.3 (see 3.5.5.1).

4.4.2.3 *Humidity test.* The blasting apparatus shall be subjected to 20 hours exposure with relative humidity between 92 and 97 percent and at air temperatures from plus 80°F. to plus 85°F., followed by 4 hours of 100 percent relative humidity with condensation at plus 75°F. to plus 80°F. Air movement and radiation negligible. At the conclusion of the humidity test, the blasting apparatus shall be inspected for evidence of damage and tested for performance in accordance with 4.4.1.1, 4.4.1.2.2 and 4.4.1.3. (See 3.5.5.3.)

4.4.3 *Operating temperature test.*

4.4.3.1 *Low ambient temperature test.* The blasting apparatus after being in an ambient air temperature of minus 35°F. to minus 40°F. unit stabilized, and while at that temperature, shall be tested for performance in accordance with 4.4.1.1, 4.4.1.2.2 and 4.4.1.3 (see 3.5.2.2).

4.4.3.2 *High ambient air temperature test.* The blasting apparatus after being in an ambient air temperature of plus 120°F. to plus 125°F. until stabilized, and while at that temperature, shall be tested for performance in accordance with 4.4.1.1, 4.4.1.2.2 and 4.4.1.3 (see 3.5.2.2).

4.4.4 *Vibration test.* The blasting apparatus shall be subjected to a vibration of simple harmonic motion for 1 hour in each of the three major axial directions at an amplitude of 0.030 inch (0.060-inch total excursion) and at a frequency variation such that the entire range between 10 and 55 cycles per second is covered each minute. At the conclusion of the test, the blasting apparatus shall be tested for performance in accordance with the applicable requirements of 4.4.1.1, 4.4.1.2.2 and 4.4.1.3. In addition for Group A, permanent change in indicator of the galvanometer shall be not greater than 0.5 percent of the readings prior to vibration (see 3.5.6.1). Evidence of damage or failure to perform satisfactorily shall be cause for rejection of the blasting apparatus. (See 3.5.6.)

4.4.5 *Shock test.* Folowing the tests specified herein, the blasting apparatus shall be inspected for evidence of damage and tested for performance in accordance with the applicable requirements of 4.4.1.1, 4.4.1.2.2 and 4.4.1.3. Evidence of damage or failure to perform satisfactorily shall be cause for rejection of the blasting apparatus. (See 3.5.7.)

4.4.5.1 *Group A.* The galvanometer shall be dropped from a height of 8 inches through each of its major axes unto a concrete surface. Permanent change in indication shall not be greater than 0.5 percent of the readings prior to being subjected to shock.

4.4.5.2 *Groups B and C.* The blasting apparatus shall be dropped from a height of 4 feet through each of its major axes unto a concrete surface.

4.4.6 *Dielectric strength test.* The blasting apparatus shall be subjected to an application of twice rated voltage applied for one minute between mutually insulated circuits and between insulated circuits and external metal parts. Evidence of damage or failure to perform satisfactorily shall be cause for rejection. (See 3.5.8.)

4.4.7 *Water submersion test (group B).* The type I blasting machine shall be submerged for 8 hours in 4 feet of water of approximately the

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same temperature as that of the machine at the time of immersion. Following, the blasting machine shall show no evidence of water seepage and shall operate in accordance with 4.4.1.2.2. Failure to perform satisfactorily shall be cause for rejection of the blasting machine. (See 3.5.9.1.)

4.5 Inspection of preparation for delivery. The preservation, packaging, packing and marking of the blasting apparatus shall be examined and tested to determine conformance with the requirements of Section 5 of this specification, and as required by specifications referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with Military Standard MIL-STD-105, with inspection Level L-7, and an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

For civil agency procurement the levels of preservation, packaging, and packing used are defined in Federal Standard No. 102.

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.* The blasting apparatus shall be cleaned in accordance with process C-1 of Military Specification MIL-P-116.

5.1.1.2 *Drying.* The blasting apparatus shall be dried in accordance with one or more of procedures D-1 through D-4 of Military Specification MIL-P-116.

5.1.1.3 *Unit packaging.* All blasting apparatus comprising one complete type and class, shall be individually preserved and packaged in accordance with Method II of Military Specification MIL-P-116.

5.1.1.3.1 *Group A and Group C* Galvanometers and Resistor assemblies shall be packaged in accordance with Method IIe of Military Specification MIL-P-116, utilizing containers conforming to Federal Specification PPP-B-566 or PPP-B-676.

5.1.1.3.2 *Group B Blasting machines (all classes)* shall be packaged in accordance with method 11b of Military Specification MIL-P-116, utilizing containers conforming to Federal Specification PPP-B-636, type I, class 1 for the inner container and type I, class 2 for the outer container.

5.1.1.4 *Intermediate package.* Group A and group C only shall be intermediate packaged six unit packages to a fiberboard container conforming to type I, grade 6, class 2, style RSC of Federal Specification PPP-B-636. Unless otherwise specified (see 6.2), the intermediate container shall be closed with 2 strips of 2-inch wide tape conforming to Federal Specification PPP-T-76. The tape shall be applied the entire length of the top and bottom seams of the box and shall extend at least 3 inches onto each end panel.

5.1.2 *Level C.* Cleaning, drying, preservation, and packaging shall be in accordance with manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* The blasting apparatus shall be packed in a container conforming to Federal Specification PPP-B-585 (class 3), PPP-B-591 (overseas type), PPP-B-601 (overseas type), PPP-B-621 (class 2), PPP-B-636 type I or II (class 3). The container shall be closed and strapped in accordance with the applicable container specification or the appendix thereto. Gross weight for fiber boxes conforming to Federal Specification PPP-B-636 shall not exceed the weight limitation of the box specification. Gross weight for all other containers shall not exceed approximately 200 pounds.

5.2.2 *Level B.* The blasting apparatus shall be packed in a container conforming to Federal Specification PPP-B-585 (class 1), PPP-B-591 (domestic type), PPP-B-601 (domestic type),

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PPP–B–621 (class 1), PPP–B–636 (class 1). Closure and weight limitations shall be as specified in paragraph 5.2.1.

5.2.3 *Level C.* The blasting apparatus packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.2.4 *Standard pack.* For civil agency procurement, the Standard Pack for this commodity shall be as specified in paragraph 5.1.1 with subparagraphs thereto and paragraph 5.2.2 (see 6.4).

5.3 Marking—civil agencies. In addition to any special marking specified in the contract or order, shipping containers shall be marked in accordance with Federal Standard No. 123.

5.4 Marking—military. In addition to any special markings required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the requirements of Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. The blasting apparatus covered by this specification is intended for use in demolitions operations. Galvanometers are used for checking the continuity of electrical circuits leading to the explosive agents. Blasting machines are used for providing sufficient current through these circuits to initiate these agents. A resistor assembly is used as a substitute for resistance present in electric caps or electric squibs, in order to test the current output of the blasting machines.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type and class required (see 1.2).

(c) Preproduction inspection when required and sample quantity if different (see 3.1).

(d) Special safety devices when required (see 3.4.9).

(e) Power requirement if different (see 3.6.4.1).

(f) Painting and finishing if different (see 3.8).

(g) Handbook of instructions if different and address where additional copies are to be sent (see 3.11).

(h) Waiver of handbook and manual approved when required (see 3.11.1).

(i) Preproduction inspection if different (see 4.2).

(j) Responsibility for preproduction inspection if different (see 4.2).

(k) Level of preservation and packaging and packing (see 5.1 and 5.2).

(l) Intermediate container closure when different (see 5.1.1.4).

6.3 *Definitions.*

6.3.1 Electric cap or squib. A device which transforms electrical energy into heat which in turn activates an explosive force strong enough to initiate the detonation of a charge of cap or squib sensitive explosives.

6.4 The Standard Pack requirements in paragraph 5.2.4 are intended for use in Stores Stock Replenishments. Procuring officers should use the Standard Pack requirements when it is known that the material will be shipped for a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue and shipment to eventual user.

6.5 Transportation descriptions and minimum weights applicable to this commodity are:

Group A

Rail: Electrical Instruments, not otherwise indexed by name.

Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

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Motor: Electrical Instruments, not otherwise, indexed.
Motor volume minimum weight 24,000 pounds.

Group B

Rail: Blasting Machines.
Carload minimum weight 30,000 pounds.
Motor: Blasting Machines.
Motor volume minimum weight 30,000 pounds.

Group C

Rail: Rheostats (resistors), or parts thereof, not otherwise indexed by name.
Carload minimum weight 30,000 pounds.
Motor: Rheostats (resistors) or parts, not otherwise indexed.
Motor volume minimum weight 30,000 pounds.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEM PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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